

This project utilizes two major platforms: TouchDesigner and Rhino Grasshopper.

Local deployment requires downloading a significant amount of proprietary software and plugins&Dazzling installing options.

I will do my best to clearly explain the prerequisites and steps involved for setting up and testing the project locally.

I. Software to be downloaded---Use the downloaded software to open the files I have uploaded to Git.

1.1-Key Software

(1)Rhino 8 (The software should come with Grasshopper included; if it doesn't, please download it.)

(2)TouchDesigner

You must use Rhino 8 to OPEN---[RhinoGH→Rhinopart.3dm/wasp---4.2.gh]

You must use TouchDesigner to OPEN---[TD→experiment.2.56]

(4.2 To use GH, you need to locate and open Grasshopper within Rhino (look for the small green icon at the top).

1.2-Key Plugins---I have uploaded the plugins and the environment to GitHub.

(1)RhinoGrasshopper---WASP(The dependency is already included in the file under the name [WASP/USER OBJECTS/WASP].)

(2)RhinoGrasshopper---RugOscGH(The dependency is already included in the file under the name [GH-TDplugin-RugOscGH].)

You can also find the link here---<https://github.com/jhorikawa/RugOscGH> to find further information.

II. How to deploy plugin--

2.1-Install WASP for Grasshopper

[This project uses WASP in Grasshopper.]

Install WASP for Grasshopper

This project uses **WASP** in Grasshopper.

Before opening the Grasshopper definition, please make sure WASP is installed correctly.

1. Locate the Grasshopper Libraries folder

Open Grasshopper, then go to:

text

File > Special Folders > Components Folder

This will open the Grasshopper components folder.

Usually the path is:

```
text
C:\Users\YOUR_USERNAME\AppData\Roaming\Grasshopper\Libraries
```

2. Copy the WASP plugin files

In this project folder, go to:

```
text
RhinoGH/wasp/
```

Inside this folder, you should see:

```
text
component folder/user objects/
```

Copy the contents as follows:

```
text
RhinoGH/wasp/component folder/
```

copy to:

```
text
C:\Users\YOUR_USERNAME\AppData\Roaming\Grasshopper\Libraries
```

Then copy:

```
text
RhinoGH/wasp/user objects/
```

to the Grasshopper user objects folder:

```
text
C:\Users\YOUR_USERNAME\AppData\Roaming\Grasshopper\UserObjects
```

You can also open this folder from Grasshopper:

```
text
File > Special Folders > User Object Folder
```

3. Unblock the files

After copying the WASP files, right-click each copied .gha or .dll file and choose:

```
text
Properties
```

If you see an option called:

```
text
Unblock
```

check it, then click:

```
text
Apply
```

Do this especially for:

```
text
.gha.dll
```

files.

4. Restart Rhino and Grasshopper

After copying and unblocking the files:

```
text
1. Close Rhino completely2. Reopen Rhino
3. Open Grasshopper
4. Open RhinoGH/wasp--4.2.gh
```

5. Check if WASP is loaded

After restarting Grasshopper, check whether WASP appears in the Grasshopper toolbar.

You should see a tab or components related to:

text

```
WASP
```

If the Grasshopper file opens without red missing components, WASP is installed correctly.

6. Open the project files

Open the files in this order:

text

```
1. RhinoGH/Rhinopart.3dm2. RhinoGH/wasp--4.2.gh
```

If everything is installed correctly, the Grasshopper definition should load normally.

Troubleshooting WASP Installation

If Grasshopper shows missing WASP components:

Problem 1: WASP files are in the wrong folder

Check again that the files from:

text

```
RhinoGH/wasp/component folder/
```

are copied into:

text

```
Grasshopper\Libraries
```

and files from:

text

```
RhinoGH/wasp/user objects/
```

are copied into:

text

```
Grasshopper\UserObjects
```

Problem 2: Files are blocked by Windows

Right-click the .gha or .dll files and check:

text

```
Properties > Unblock > Apply
```

Then restart Rhino.

Problem 3: Rhino was not restarted

Grasshopper only loads plugins when Rhino starts.
After installing WASP, fully close and reopen Rhino.

Problem 4: Wrong Rhino version

If the WASP components still do not load, check that the project is opened in a compatible Rhino / Grasshopper version.

2.2-Deploy -RugOscGH

RugOscGH Installation

This project uses **RugOscGH** to send OSC data between TouchDesigner and Grasshopper.

Before opening the Grasshopper definition, please install the RugOscGH plugin.

1. Locate the Grasshopper Libraries folder

Open Grasshopper, then go to:

text
File > Special Folders > Components Folder

This will open the Grasshopper Libraries folder.

The folder is usually located here:

text
C:\Users\YOUR_USERNAME\AppData\Roaming\Grasshopper\Libraries

2. Copy RugOscGH files

In this project folder, go to:

text
GH-TDplugin-RugOscGH/

Copy these two files:

text
RugOscGH.ghaRug.Osc.dll

Paste them into the Grasshopper Libraries folder:

text
C:\Users\YOUR_USERNAME\AppData\Roaming\Grasshopper\Libraries

Important: copy the .gha and .dll files directly into Libraries.

Do **not** only copy the source-code folder RugOscGH-master.

3. Unblock the files

After copying the files, right-click each file:

text
RugOscGH.ghaRug.Osc.dll

Choose:

text
Properties

If you see an option called:

text
Unblock

check it, then click:

text
Apply

This step is important because Windows may block downloaded .gha or .dll files.

4. Restart Rhino and Grasshopper

After copying and unblocking the files:

text
1. Close Rhino completely2. Reopen Rhino
3. Open Grasshopper

Grasshopper loads plugins only when Rhino starts, so restarting Rhino is necessary.

5. Check if RugOscGH is loaded

After restarting Grasshopper, check the Grasshopper toolbar.

You should see RugOscGH-related OSC components, such as:

text
OSC SenderOSC Receiver
OSC Dispatch

The exact component names may vary, but the Grasshopper file should open without missing RugOscGH components.

6. Open the project Grasshopper file

After RugOscGH is installed, open:

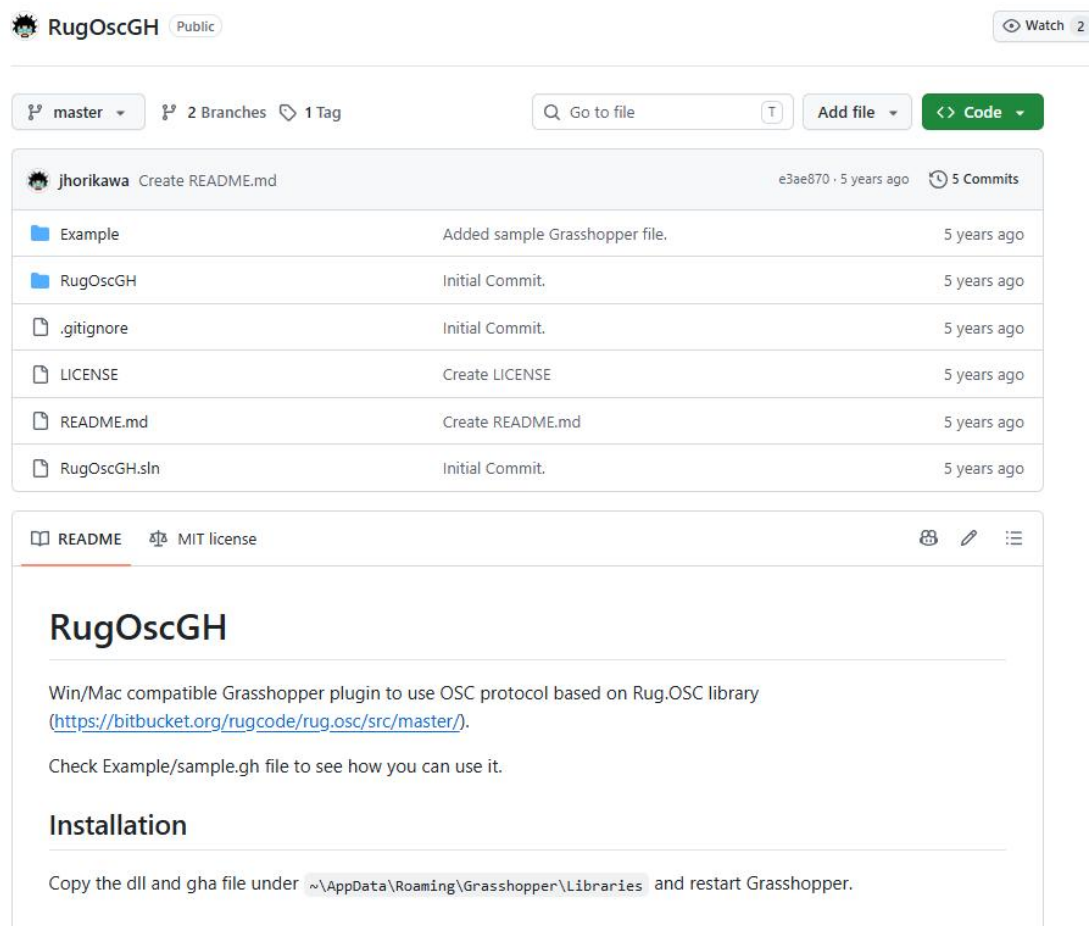
```
text
RhinoGH/Rhinopart.3dm
```

Then open:

```
text
RhinoGH/wasp--4.2.gh
```

If the file opens without red missing RugOscGH components, the plugin is installed correctly.

MORE INFORMATION CAN BE FOUND HERE



The screenshot shows the GitHub repository page for **RugOscGH** by user **jhorikawa**. The repository is public and has 2 branches and 1 tag. The commit history shows several files added, including `Example`, `RugOscGH`, `.gitignore`, `LICENSE`, `README.md`, and `RugOscGH.sln`. The README file is selected, showing the title **RugOscGH** and the MIT license. The README content describes the plugin as a Win/Mac compatible Grasshopper plugin using the OSC protocol based on the Rug.OSC library. It provides a link to the Bitbucket source code and instructions for installation, including copying the DLL and gha files to the Grasshopper Libraries folder.

RugOscGH

Win/Mac compatible Grasshopper plugin to use OSC protocol based on Rug.OSC library (<https://bitbucket.org/rugcode/rug.osc/src/master/>).

Check Example/sample.gh file to see how you can use it.

Installation

Copy the dll and gha file under `~\AppData\Roaming\Grasshopper\Libraries` and restart Grasshopper.

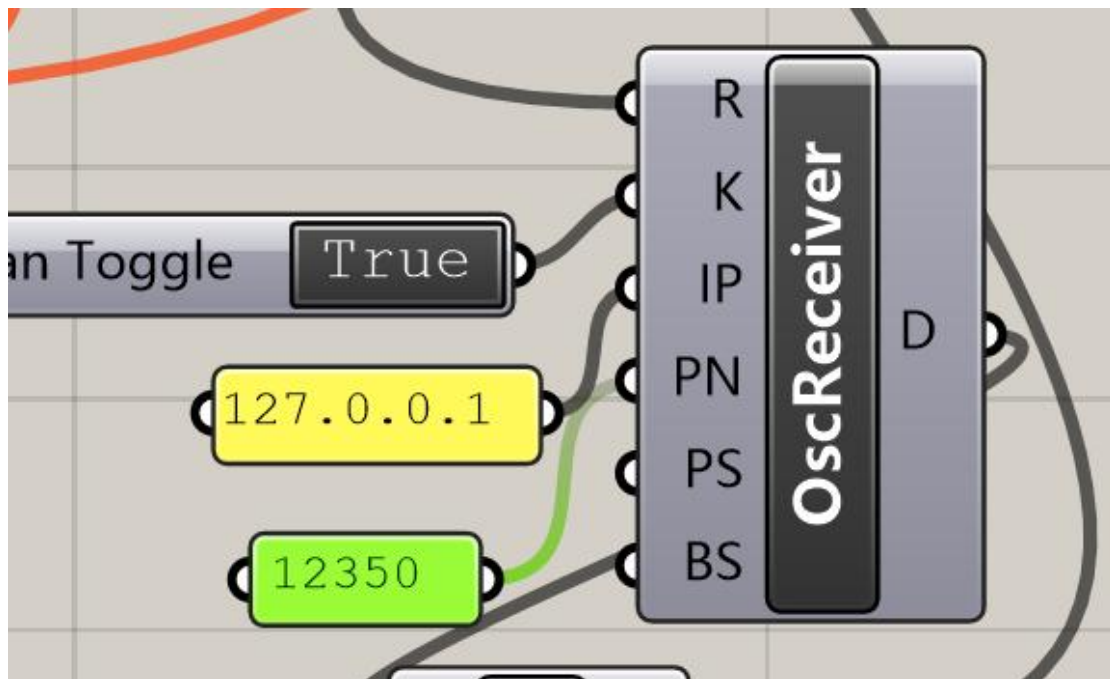
III. start the whole project--

3.1-If everything goes smoothly, you must open the files `Rhinoart.3dm` (using Rhino 8) and `wasp--4.2.gh` (using Grasshopper within Rhino).

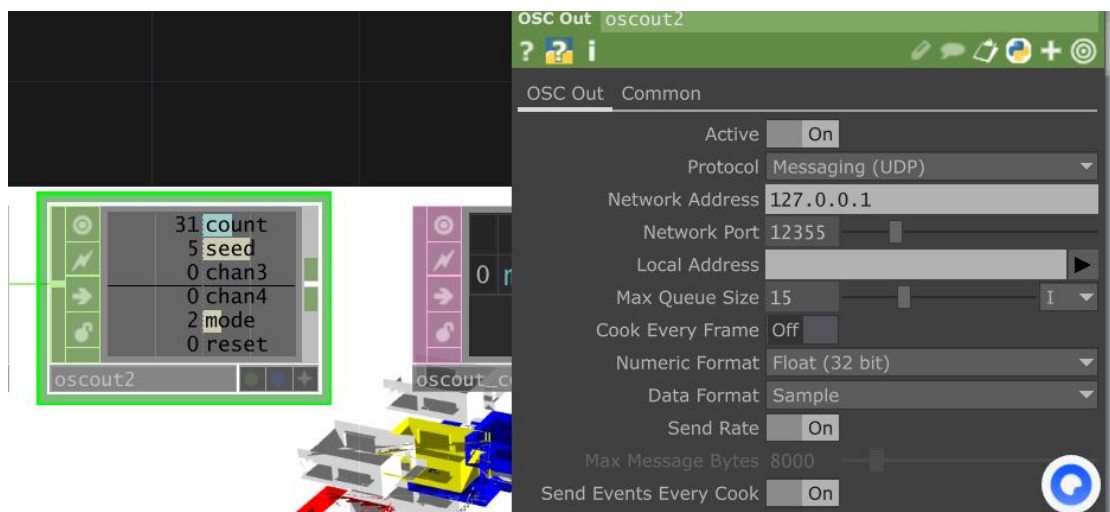
 Rhinoart.3dm	2026/9/5 2:12	Rhino 3-D Model	3,849 KB
 wasp--4.2.gh	2026/9/5 2:12	Grasshopper De...	193 KB

Then open td file  experiment.2.56.toe 2026/9/2 13:39 TouchDesigner ... 36 KB

After opening everything,IN GRASSHOPPER WASP 4.2,find osc reciever,Change the number of the yellow node connected to pn to a random number (preferably one greater than 12356).



IN TD,Locate the "Oscout2" node in the TD workflow and set its port to the same as grasshopper osc reciever



Note: If the workflow is functioning correctly but the game data fails to refresh, you can try

changing the port number to resolve the issue; however, the port numbers in both TD and Grasshopper must match.

Then ,switch TD to Perform Mode. Generally, simply pressing "Start" while in Perform Mode or Work Mode is sufficient to refresh successfully.

If everything has gone smoothly up to this point... then the program should be ready to use.

If the local deployment fails, I apologize for not explaining the process clearly enough; you might try asking GPT for help troubleshooting. Also, I failed to consider in time the need to package this as a readily downloadable application.

Additionally, my demo video proves that the program actually works; the link is below: If everything has gone smoothly up to this point... then the program should be ready to use.

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Additionally, my demo video proves that the program actually works; the link is below:

<https://youtu.be/gn15r1sdw-c>